UC SAN DIEGO / ECE 251C

6
Generalized sampling and multiresolution representations

Acquisition kernels, local averages, and Haar multiresolution

What information is added when we double the resolution of a signal?

6.1 Generalized sampling

We now connect local averages to a sampling device and organize their information by resolution. The domain is the whole line $\mathbb{R}$, with no periodic extension.

Recall: ideal sampling. With unit sampling step,

$$f[n] = f(n) = \langle f, \delta(\cdot - n)\rangle, \qquad n \in \mathbb{Z}.$$

The impulse pairing means evaluation at n, whenever that value is defined; it is not an L^2 inner product with δ.

Generalized sampling. For $f \in L^2(\mathbb{R})$, replace the impulse by a real-valued acquisition function $\phi \in L^2(\mathbb{R})$:

$$a[n] := \langle f, \phi(\cdot - n)\rangle_{L^2} = \int_{\mathbb{R}} f(t)\phi(t-n)\,\mathrm{d}t. \tag{6.1}$$

As before, conjugate the second function; here $\phi^* = \phi$ because ϕ is real. This measurement is an integral, not a single function value.

Filtering before sampling. Set $\widetilde{\phi}(t) := \phi(-t)$. Then

$$a[n] = \int_{\mathbb{R}} f(t)\widetilde{\phi}(n-t)\,\mathrm{d}t = (f * \widetilde{\phi})(n).$$

The impulse response is $\widetilde{\phi}$: first filter, then sample the output.

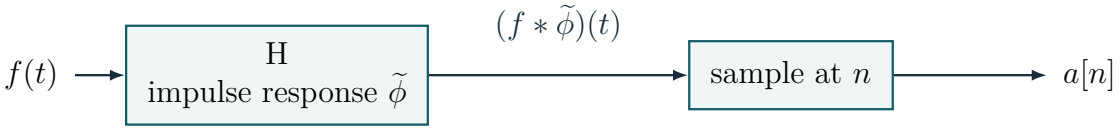


Figure 6.1: Generalized sampling is filtering followed by ideal sampling. The reflection in $\widetilde{\phi}(t) = \phi(-t)$ converts the inner-product formula into convolution.

6.2 Box measurements and piecewise-constant reconstruction

The simplest acquisition function is the unit box:

$$\phi(t) = \mathbf{1}_{[0,1)}(t), \qquad \tilde{\phi}(t) = \phi(-t).$$

The choice of values at the endpoints does not affect an integral.

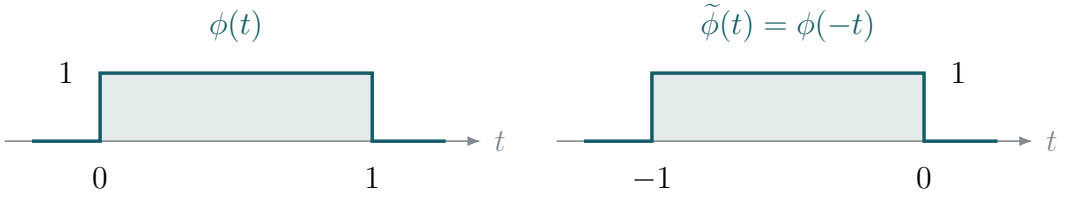


Figure 6.2: The acquisition box and its reflected impulse response. At output location n, convolution with the reflected box integrates f over $[n, n+1)$.

For this choice, equation (6.1) becomes

$$a[n] = \int_n^{n+1} f(t)\,\mathrm{d}t. \tag{6.2}$$

The interval has length one, so the integral is also its average. The translates $\phi(t-n)$ have disjoint interiors and energy one. Consequently they are orthonormal.

Reconstruct using these same boxes:

$$f_0(t) \coloneqq \sum_{n\in\mathbb{Z}} a[n]\phi(t-n). \tag{6.3}$$

For $n \le t < n+1$, only the nth term is nonzero. Thus $f_0(t) = a[n]$: we replace each unit interval by its average value.

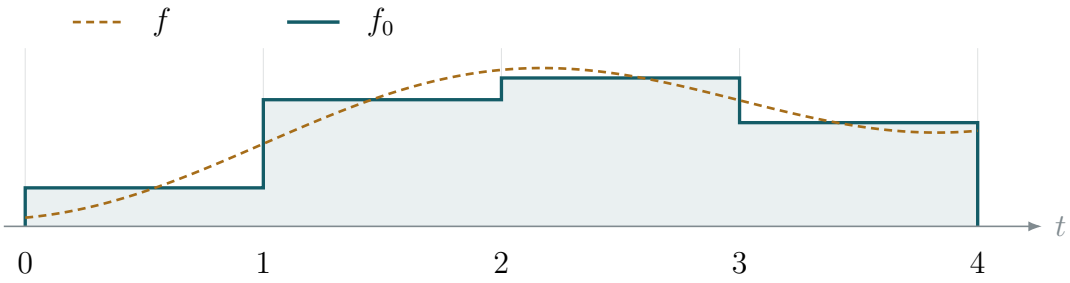


Figure 6.3: Four distinct unit-cell averages of $f(t) = 0.7 + 0.25t - 0.6\cos(\pi t/2)$ on $[0, 4)$, with zero values outside that interval. Each box height is an average, not a point sample.

The acquisition and reconstruction functions have different roles. The reflected box forms the measurement by convolution; the unreflected box places the measured

average back on the interval $[n, n+1)$. A single set of unit-width averages does not determine an arbitrary signal exactly.

6.3 Changing resolution without changing energy

To double the resolution, replace $\phi(t-n)$ by $\phi(2t-n)$. Its nonzero interval has half the width. More generally, for an integer resolution i, define

$$I_{i,n} := [n2^{-i}, (n+1)2^{-i}), \qquad n \in \mathbb{Z}.$$

The function $\phi(2^i t - n)$ occupies this cell, whose width is 2^{-i}. Increasing i means finer resolution; negative i gives cells wider than one.

The amplitude needs a normalization. The unnormalized box has energy

$$\int_{\mathbb{R}} |\phi(2^i t - n)|^2 \, \mathrm{d}t = 2^{-i} \int_{\mathbb{R}} |\phi(u)|^2 \, \mathrm{d}u = 2^{-i}, \qquad u = 2^i t - n.$$

Its energy shrinks with its support. Normalize all sampling functions to unit energy:

$$\boxed{\phi_{i,n}(t) := 2^{i/2}\phi(2^i t - n), \qquad i, n \in \mathbb{Z}.} \tag{6.4}$$

With $u = 2^i t - n$ and $\mathrm{d}t = 2^{-i}\mathrm{d}u$,

$$\begin{aligned} \|\phi_{i,n}\|_{L^2}^2 &= 2^i \int_{\mathbb{R}} |\phi(2^i t - n)|^2 \, \mathrm{d}t \\ &= 2^i 2^{-i} \int_{\mathbb{R}} |\phi(u)|^2 \, \mathrm{d}u = 1. \end{aligned}$$

The factor $2^{i/2}$ compensates exactly for the change in width.

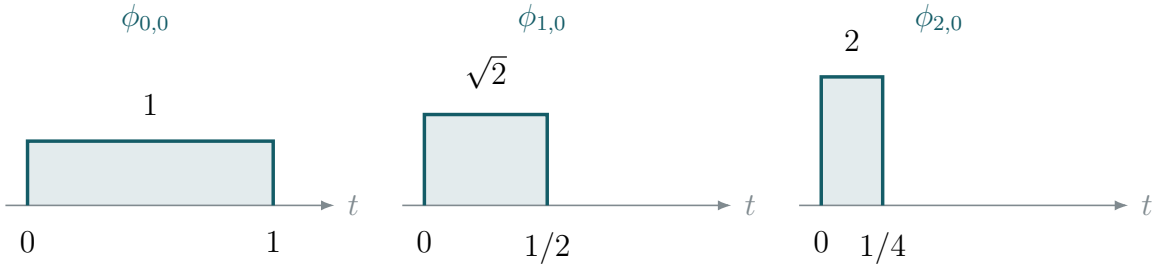


Figure 6.4: Unit-energy boxes at three resolutions, drawn with common horizontal and vertical units. Halving the width multiplies the height by $\sqrt{2}$, keeping height squared times width equal to one.

Unit *energy* is not unit *area*: $\int_{\mathbb{R}} \phi_{i,n}(t) \, \mathrm{d}t = 2^{-i/2}$. This distinction determines how a measured coefficient becomes a reconstructed signal value.

6.4 Coefficients and local averages

At resolution i, acquire the normalized coefficients

$$a_i[n] \coloneqq \langle f, \phi_{i,n} \rangle_{L^2} = 2^{i/2} \int_{I_{i,n}} f(u)\, \mathrm{d}u. \tag{6.5}$$

Reconstruction uses the same normalized functions:

$$f_i(t) \coloneqq \sum_{n \in \mathbb{Z}} a_i[n] \phi_{i,n}(t).$$

If $t \in I_{i,n}$, the only surviving factor is $\phi_{i,n}(t) = 2^{i/2}$. Therefore

$$\boxed{f_i(t) = 2^{i/2} a_i[n] = 2^i \int_{I_{i,n}} f(u)\, \mathrm{d}u, \qquad t \in I_{i,n}.} \tag{6.6}$$

Since $|I_{i,n}| = 2^{-i}$, the last expression is precisely the cell average.

Three quantities, not one. If $m_i[n]$ is the average value on a cell, then

$$\underbrace{\int_{I_{i,n}} f(u)\, \mathrm{d}u}_{\text{raw integral}} = 2^{-i} m_i[n], \qquad \underbrace{a_i[n]}_{\text{unit-energy coefficient}} = 2^{-i/2} m_i[n].$$

Multiplication by the reconstruction height $2^{i/2}$ returns $m_i[n]$. In particular, any cell on which f is constant with value b is reconstructed with that same value. Cells crossing a change in f may have a different average.

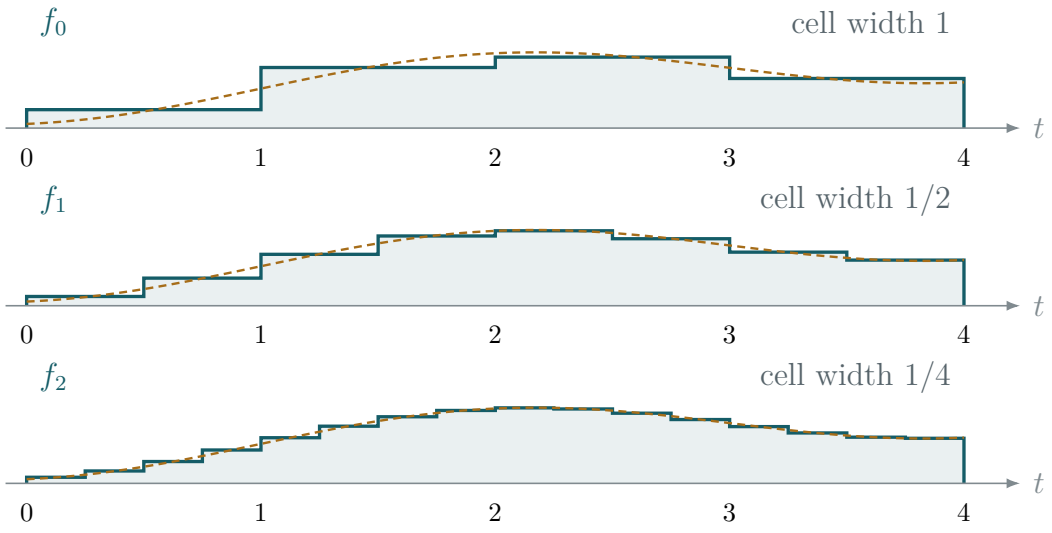


Figure 6.5: The same signal and averaging rule at successively finer resolutions. The plotted step heights are averages; their normalized expansion coefficients are smaller by the factor $2^{-i/2}$.

Finer cells retain more variation within each unit interval. The averaging rule is unchanged; only the scale changes.

6.5 The detail between two resolutions

Consider one cell $I_{i,n}$, split into two children. Suppose the finer approximation has value α on the left child and β on the right. Its coarse average is

$$m = \frac{\alpha + \beta}{2}.$$

Subtracting the average gives two opposite residuals:

$$\alpha - m = \frac{\alpha - \beta}{2}, \qquad \beta - m = -\frac{\alpha - \beta}{2}.$$

Thus a finer approximation is a coarse average plus a local difference.

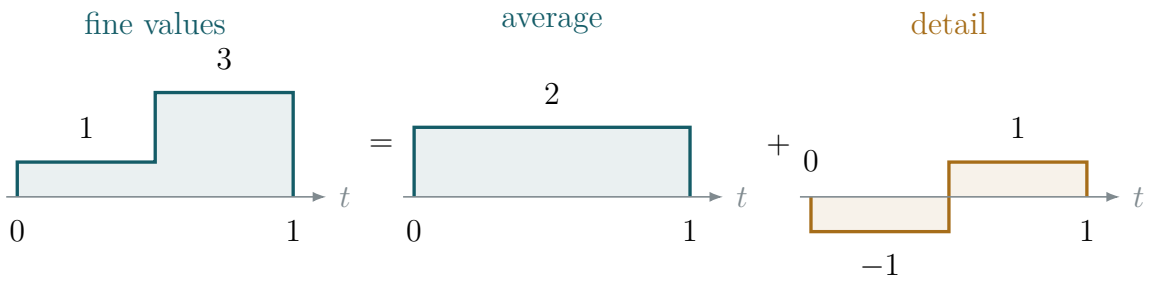


Figure 6.6: Two fine values, 1 and 3, become the average 2 and opposite details $-1, +1$. Adding the two right-hand plots returns the left-hand plot exactly.

The Haar wavelet. The pattern that describes this difference is

$$\psi(t) := \mathbf{1}_{[0,1/2)}(t) - \mathbf{1}_{[1/2,1)}(t), \qquad \psi_{i,n}(t) := 2^{i/2}\psi(2^i t - n). \tag{6.7}$$

The function ψ is the Haar *mother wavelet*; ϕ is the Haar *scaling function*, also called the father wavelet. Direct calculation gives

$$\int_{\mathbb{R}} \psi(t)\,\mathrm{d}t = \frac{1}{2} - \frac{1}{2} = 0, \qquad \|\psi\|_{L^2}^2 = \frac{1}{2} + \frac{1}{2} = 1.$$

The substitution from equation (6.4) also gives $\|\psi_{i,n}\|_{L^2} = 1$ and $\int_{\mathbb{R}} \psi_{i,n}(t)\,\mathrm{d}t = 0$.

Define the detail function $d_i := f_{i+1} - f_i$. On the chosen parent cell,

$$d_i(t) = \frac{\alpha - \beta}{2}\,\psi(2^i t - n) = 2^{-i/2}\frac{\alpha - \beta}{2}\,\psi_{i,n}(t). \tag{6.8}$$

Equal child averages give zero detail. These are changes relative to a resolution, not isolated Fourier frequencies.

6.6 A complete multiresolution decomposition

Take a signal supported on $[0, 1)$ with successive eighth-cell values

$$(1, 3, 5, 7, 7, 7, 5, 5).$$

It is already constant on each cell at resolution 3, so $f = f_3$. Repeated averaging gives

resolution	cell width	approximation values
3	1/8	$(1, 3, 5, 7, 7, 7, 5, 5)$
2	1/4	$(2, 6, 7, 5)$
1	1/2	$(4, 6)$
0	1	(5)

For example, $(1 + 3)/2 = 2$, $(5 + 7)/2 = 6$, and $(2 + 6)/2 = 4$. The difference at each stage records exactly what the averaging removes.

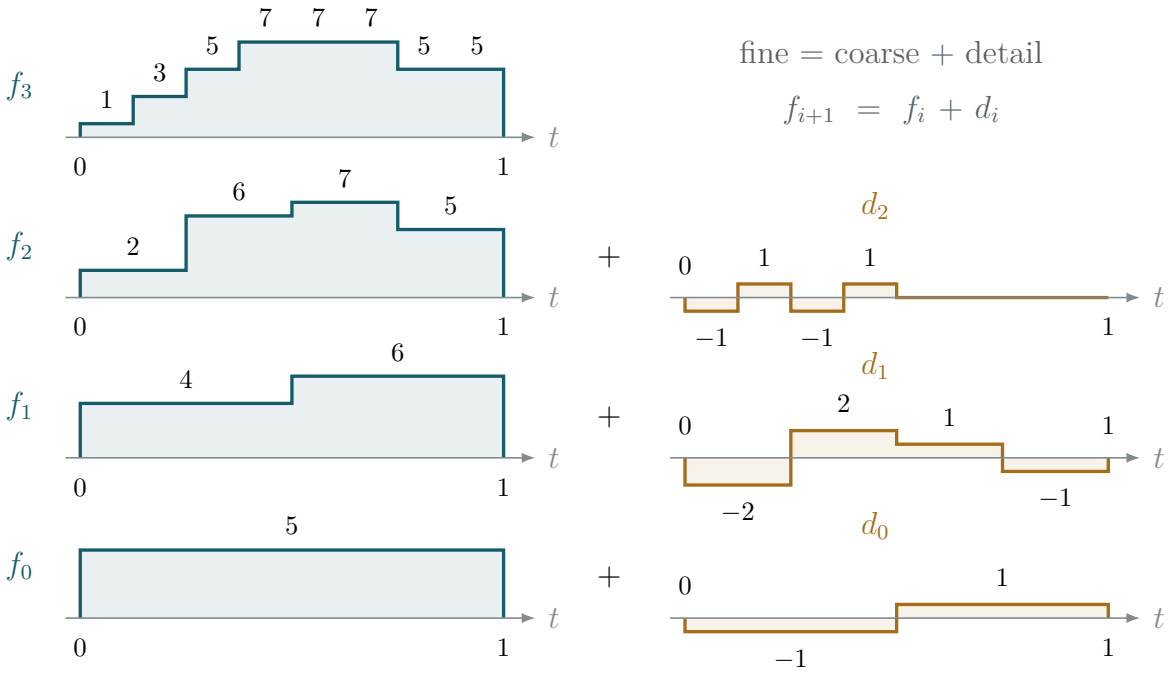


Figure 6.7: A three-level decomposition. All plots use the same vertical scale and are zero outside $[0, 1)$. Each detail has zero average on every parent cell.

Read each row as $f_{i+1} = f_i + d_i$. Repeated substitution gives

$$\boxed{f_3 = f_0 + d_0 + d_1 + d_2.} \tag{6.9}$$

The numbers in the plots are *function values*, not normalized basis coefficients. We calculate those coefficients after deriving the two-scale equations.

6.7 Approximation spaces and detail spaces

At a fixed resolution i, define

$$V_i := \overline{\operatorname{span}\{\phi_{i,n} : n \in \mathbb{Z}\}}^{L^2}, \tag{6.10}$$

$$W_i := \overline{\operatorname{span}\{\psi_{i,n} : n \in \mathbb{Z}\}}^{L^2}. \tag{6.11}$$

The overline means *closure in the L^2 norm*: include not only finite linear combinations, but also signals that these combinations approximate arbitrarily well in energy. This allows finite-energy signals extending across infinitely many cells.

The space V_i consists of finite-energy functions constant on each $I_{i,n}$. The space W_i consists of finite-energy functions constant on each half-cell, with opposite values on the two halves of every parent cell.

Orthonormal functions at each resolution. Distinct cells have disjoint interiors and all the generators have unit energy. Hence

$$\langle \phi_{i,n}, \phi_{i,m} \rangle_{L^2} = \langle \psi_{i,n}, \psi_{i,m} \rangle_{L^2} = \delta[n-m].$$

Also $V_i \perp W_i$. For different cells the inner product vanishes by disjointness; on the same cell,

$$\langle \phi_{i,n}, \psi_{i,n} \rangle_{L^2} = 2^i \left(2^{-i-1} - 2^{-i-1}\right) = 0.$$

The notation $V_i \perp W_i$ means that every element of one space is orthogonal to every element of the other.

The cell-average approximation is a projection. Write $P_i f = f_i$. On each cell, equation (6.6) gives

$$\int_{I_{i,n}} (f(t) - f_i(t))\, \mathrm{d}t = 0.$$

Completing the square on a cell gives $\int_I |f-m|^2 = \int_I |f|^2 - |I||m|^2 \geq 0$, where m is its mean. Summing over cells shows that f_i has finite energy. The zero residual integrals make $f - f_i$ orthogonal to V_i. Thus, for $v \in V_i$,

$$\|f - v\|_{L^2}^2 = \|f - f_i\|_{L^2}^2 + \|f_i - v\|_{L^2}^2. \tag{6.12}$$

Therefore $P_i f$ is the unique best approximation in V_i. In particular,

$$\|f\|_{L^2}^2 = \|f - P_i f\|_{L^2}^2 + \|P_i f\|_{L^2}^2, \qquad \|P_i f\|_{L^2} \leq \|f\|_{L^2}.$$

For an arbitrary acquisition function, reconstruction with the same translates need not be a projection. Here it is a projection because the Haar boxes are orthonormal.

6.8 Nested spaces and the two-scale equations

A function constant on a cell is also constant on each of its two children. A Haar detail also has constant values on those children. Consequently

$$V_i \subset V_{i+1}, \qquad W_i \subset V_{i+1}.$$

In particular, both $\phi \in V_0$ and $\psi \in W_0$ can be built from the finer boxes in V_1.

Split the box. The two half-intervals partition $[0, 1)$, so

$$\phi(t) = \phi(2t) + \phi(2t-1).$$

Since $\phi_{1,0}(t) = \sqrt{2}\phi(2t)$ and $\phi_{1,1}(t) = \sqrt{2}\phi(2t-1)$,

$$\boxed{\phi(t) = \frac{1}{\sqrt{2}}\phi_{1,0}(t) + \frac{1}{\sqrt{2}}\phi_{1,1}(t).} \tag{6.13}$$

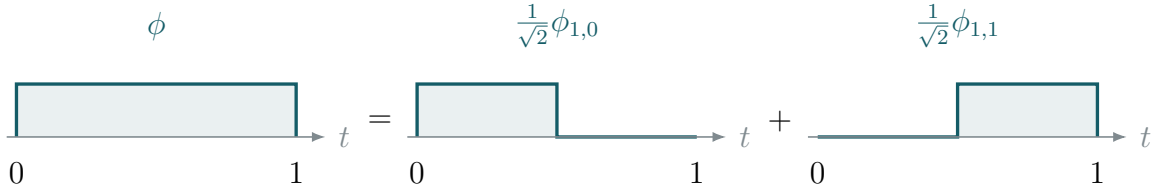


Figure 6.8: The scaling function is the sum of two weighted finer boxes. Each weighted term has height one on its own half-interval.

Split the wavelet. The right half has the opposite sign:

$$\psi(t) = \phi(2t) - \phi(2t-1).$$

Therefore

$$\boxed{\psi(t) = \frac{1}{\sqrt{2}}\phi_{1,0}(t) - \frac{1}{\sqrt{2}}\phi_{1,1}(t).} \tag{6.14}$$

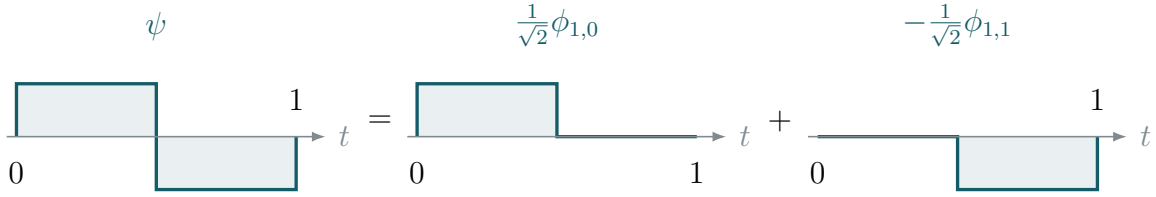


Figure 6.9: The wavelet uses the same two finer boxes with opposite coefficients. The negative term supplies the right-hand half of the Haar pattern.

These are the *two-scale equations*. Each expresses a function at one scale in terms of scaling functions at the next finer scale. The factors $1/\sqrt{2}$ are basis coefficients: they turn height-$\sqrt{2}$ boxes into height-one pieces.

6.9 The two-scale coefficients are filters

Write the equations in their filter form:

$$\phi(t) = \sum_{n\in\mathbb{Z}} g_0[n]\sqrt{2}\phi(2t-n), \tag{6.15}$$

$$\psi(t) = \sum_{n\in\mathbb{Z}} g_1[n]\sqrt{2}\phi(2t-n), \tag{6.16}$$

where

$$g_0[n] = \frac{\delta[n]+\delta[n-1]}{\sqrt{2}}, \qquad g_1[n] = \frac{\delta[n]-\delta[n-1]}{\sqrt{2}}.$$

Only the taps at $n = 0, 1$ are nonzero.

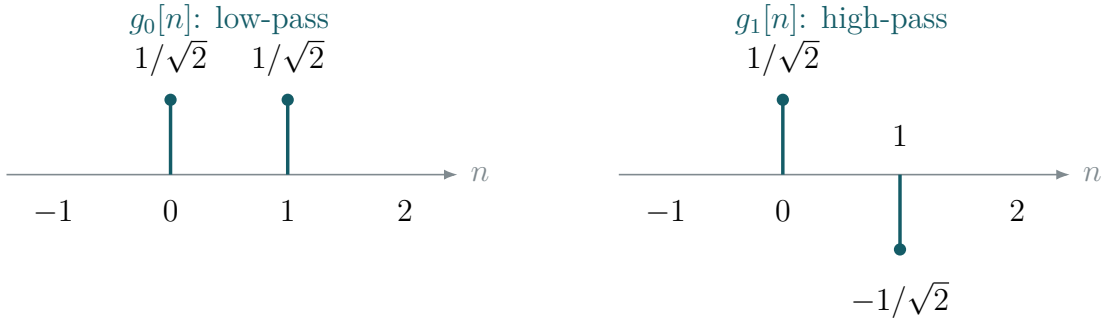


Figure 6.10: Both masks have energy one; their inner product is $1/2 - 1/2 = 0$.

Frequency responses (ω in radians per sample):

$$G_0(\mathrm{e}^{\mathrm{j}\omega}) = \frac{1+\mathrm{e}^{-\mathrm{j}\omega}}{\sqrt{2}} = \sqrt{2}\mathrm{e}^{-\mathrm{j}\omega/2}\cos(\omega/2),$$

$$G_1(\mathrm{e}^{\mathrm{j}\omega}) = \frac{1-\mathrm{e}^{-\mathrm{j}\omega}}{\sqrt{2}} = \sqrt{2}\mathrm{j}\mathrm{e}^{-\mathrm{j}\omega/2}\sin(\omega/2).$$

The factorization is $1 \pm \mathrm{e}^{-\mathrm{j}\omega} = \mathrm{e}^{-\mathrm{j}\omega/2}(\mathrm{e}^{\mathrm{j}\omega/2} \pm \mathrm{e}^{-\mathrm{j}\omega/2})$. The sum in parentheses is $2\cos(\omega/2)$; the difference is $2\mathrm{j}\sin(\omega/2)$.

At $\omega = 0, \pi$, respectively, the responses are $(\sqrt{2}, 0)$ for g_0 and $(0, \sqrt{2})$ for g_1: an average rejects alternating signs, while a difference rejects constants.

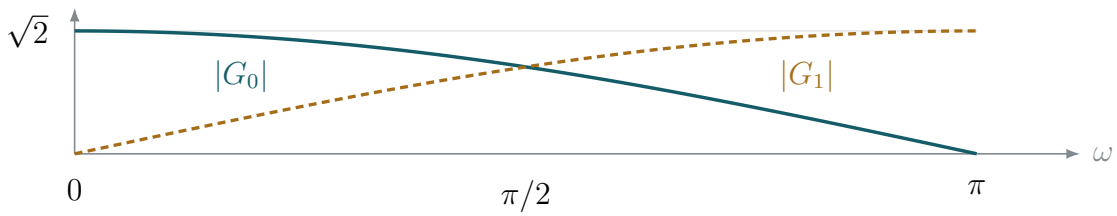


Figure 6.11: Magnitude responses on $[0, \pi]$. Their squared magnitudes add to two; neither filter has an ideal rectangular passband.

6.10 An orthogonal direct sum

Proposition 6.1. At every integer resolution,

$$\boxed{V_{i+1} = V_i \oplus W_i.}$$

Every $v \in V_{i+1}$ has a unique decomposition $v = p + q$, with $p \in V_i$ and $q \in W_i$, and the two components are orthogonal.

Proof. On each parent cell, let α, β be the two values of v. Set

$$m = \frac{\alpha + \beta}{2}, \qquad r = \frac{\alpha - \beta}{2}.$$

Let $p = m$ on the whole parent cell, and let q take the values $r, -r$ on its two halves. Then $p \in V_i$, $q \in W_i$, and

$$m + r = \alpha, \qquad m - r = \beta.$$

These functions have finite energy: the identity

$$|m|^2 + |r|^2 = \frac{|\alpha|^2 + |\beta|^2}{2}$$

shows that their energies add to the energy of v after multiplication by the parent-cell width and summation over cells.

We already proved $V_i \perp W_i$. If a function u belongs to both spaces, it is orthogonal to itself, so $\|u\|_{L^2}^2 = 0$. Thus $V_i \cap W_i = \{0\}$. Subtracting two decompositions would put their difference in this intersection, proving uniqueness. □

The symbol $\oplus$ records uniqueness; here the sum is also orthogonal. For the approximations of a fixed signal,

$$P_{i+1} f = P_i f + d_i, \qquad d_i = P_{i+1} f - P_i f \in W_i. \tag{6.17}$$

Moreover, d_i is the orthogonal projection of f onto W_i. Indeed, $f - d_i = (f - P_{i+1} f) + P_i f$: the first term is orthogonal to V_{i+1}, and the second to W_i.

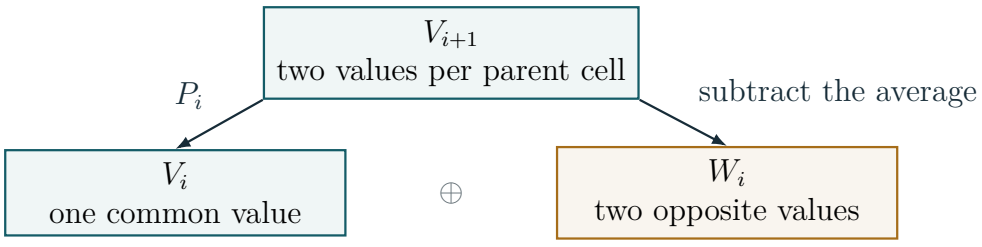


Figure 6.12: Averages and details occupy complementary, orthogonal subspaces of the next finer approximation space.

Compute the coefficients without resampling the function. Define

$$a_i[n] = \langle f, \phi_{i,n} \rangle_{L^2}, \qquad b_i[n] \coloneqq \langle f, \psi_{i,n} \rangle_{L^2}.$$

Dilating and translating equations (6.13) and (6.14) gives

$$\phi_{i,n} = \frac{\phi_{i+1,2n} + \phi_{i+1,2n+1}}{\sqrt{2}},$$
$$\psi_{i,n} = \frac{\phi_{i+1,2n} - \phi_{i+1,2n+1}}{\sqrt{2}}.$$

For example, substitute $2^i t - n$ for t in the box equation and multiply by $2^{i/2}$; each child has normalization $2^{(i+1)/2}$. Taking inner products, the coefficients $1/\sqrt{2}$ are real, so conjugation does not change them:

$$\boxed{\begin{aligned} a_i[n] &= \frac{a_{i+1}[2n] + a_{i+1}[2n+1]}{\sqrt{2}}, \\ b_i[n] &= \frac{a_{i+1}[2n] - a_{i+1}[2n+1]}{\sqrt{2}}. \end{aligned}} \tag{6.18}$$

Adding and subtracting these equations recovers the fine coefficients:

$$a_{i+1}[2n] = \frac{a_i[n] + b_i[n]}{\sqrt{2}}, \qquad a_{i+1}[2n+1] = \frac{a_i[n] - b_i[n]}{\sqrt{2}}. \tag{6.19}$$

The detail reconstructs as $d_i = \sum_n b_i[n]\psi_{i,n}$. Keeping both coefficient outputs gives perfect reconstruction.

Return to the eight-cell example. A cell value at resolution i must be multiplied by $2^{-i/2}$ to obtain its coefficient. Applying equation (6.18),

i	a_i	b_i
3	$(1,3,5,7,7,7,5,5)/\sqrt{8}$	—
2	$(1, 3, 7/2, 5/2)$	$(-1/2, -1/2, 0, 0)$
1	$(2\sqrt{2}, 3\sqrt{2})$	$(-\sqrt{2}, 1/\sqrt{2})$
0	(5)	(-1)

For the first pair, $\sqrt{8}\sqrt{2} = 4$, so $a_2[0] = (1+3)/4 = 1$ and $b_2[0] = (1-3)/4 = -1/2$. The complete Haar expansion is

$$f = 5\phi - \psi - \sqrt{2}\psi_{1,0} + \frac{1}{\sqrt{2}}\psi_{1,1} - \frac{1}{2}\psi_{2,0} - \frac{1}{2}\psi_{2,1}. \tag{6.20}$$

All other coefficients at these resolutions are zero. The filter operations rearrange the same information; discarding details is the information-losing step.

6.11 The Haar multiresolution analysis

The approximation spaces form a nested family:

$$\cdots \subset V_{-2} \subset V_{-1} \subset V_0 \subset V_1 \subset V_2 \subset \cdots \subset L^2(\mathbb{R}).$$

Each is a dilated version of V_0: $v \in V_i$ exactly when $v(2^{-i}\,\cdot) \in V_0$. All grids use the same origin, with finer cells as i increases.

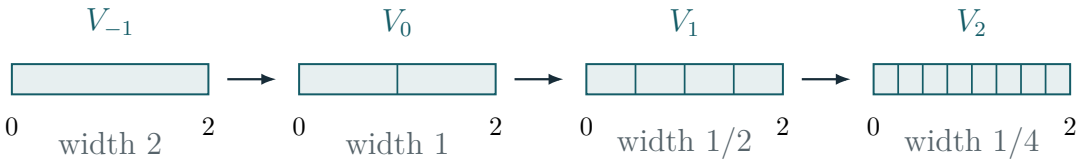


Figure 6.13: The same displayed interval, $[0, 2)$, at four resolutions. The spaces themselves are defined on the entire real line; only the cell width changes.

The two limiting statements in the multiresolution picture have precise meanings:

$$\boxed{\overline{\bigcup_{i\in\mathbb{Z}} V_i}^{\,L^2} = L^2(\mathbb{R}), \qquad \bigcap_{i\in\mathbb{Z}} V_i = \{0\}.} \tag{6.21}$$

The first says that sufficiently fine piecewise-constant approximations can achieve arbitrarily small energy error. The second says that no nonzero finite-energy signal can be constant on the cells at *every* scale.

Finer resolutions recover the signal. For a signal p continuous on a closed, bounded interval, averaging over a cell I cannot change it by more than its variation inside that cell:

$$\left| p(t) - \frac{1}{|I|}\int_I p(u)\,\mathrm{d}u \right| \le \sup_{u\in I} |p(t) - p(u)|, \qquad t \in I.$$

As the cell width tends to zero, this bound tends to zero uniformly on that interval. Hence the energy error tends to zero as well.

To pass to a general $f \in L^2(\mathbb{R})$, first discard tails of arbitrarily small energy. On each of the remaining finitely many unit intervals, approximate f by a finite Fourier sum, using the Fourier-series theorem from Lecture 5. Call the resulting piecewise-smooth signal p; its support is bounded. Fine dyadic cells do not cross the integer boundaries, so the preceding argument applies on each interval. Since P_i cannot increase the L^2 norm,

$$\|f - P_i f\|_{L^2} \le 2\|f - p\|_{L^2} + \|p - P_i p\|_{L^2}.$$

First make $\|f - p\|_{L^2}$ arbitrarily small, then take i large enough to make the last term small. Thus $\|f - P_i f\|_{L^2} \to 0$, proving the first part of equation (6.21).

Coarser resolutions leave no finite-energy background. Suppose v belongs to every V_i. At resolution $i = -m$, it is constant on $[0, 2^m)$ and on $[-2^m, 0)$. As m grows, these intervals exhaust the two half-lines. Their overlaps force the constants to agree on each half-line. A nonzero constant on a half-line has infinite energy. Hence both constants must be zero, proving the second part of equation (6.21).

The coarse projections themselves also tend to zero. For a finite-energy signal p with bounded support, write $L = 2^{-i}$. Its cell averages satisfy

$$\begin{aligned} \|P_i p\|_{L^2}^2 &= \frac{1}{L} \sum_n \left| \int_{I_{i,n}} p(t)\, \mathrm{d}t \right|^2 \\ &\le \frac{1}{L} \left(\int_{\mathbb{R}} |p(t)|\, \mathrm{d}t \right)^2 \longrightarrow 0 \qquad (i \to -\infty). \end{aligned}$$

Here the inequality follows from $|\int_I p| \le \int_I |p|$ and $\sum_n u_n^2 \le (\sum_n u_n)^2$ for $u_n \ge 0$. For general f, approximate by such a p and use

$$\|P_i f\|_{L^2} \le \|P_i(f-p)\|_{L^2} + \|P_i p\|_{L^2} \le \|f-p\|_{L^2} + \|P_i p\|_{L^2}.$$

The whole-line setting matters: unlike on $[0,1]$, the constant function is not a nonzero finite-energy background on $\mathbb{R}$.

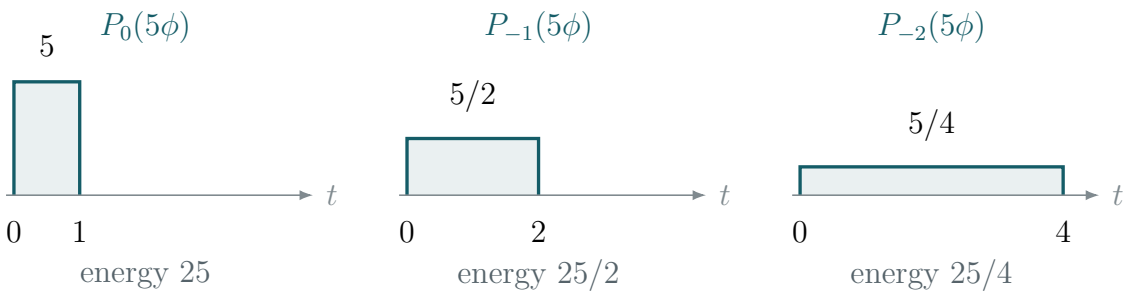


Figure 6.14: Coarsening the example's background 5ϕ. Its integral stays equal to five, but its amplitude and energy decrease as that integral is spread over longer intervals.

Together, the nested scales, the two limits, and the orthonormal translates of ϕ give the *Haar multiresolution analysis*, abbreviated MRA. It is generated by one scaling function and one rule for changing scale.